The Silent Gap in Sepsis Care

"A ward patient begins to deteriorate at 2:14 PM. Vital signs worsen. Lactate rises. Antibiotics are delayed. No documented senior review occurs. By evening, the patient develops septic shock."

PROBLEM STATEMENT

In hospital wards, patients with sepsis frequently show early physiological decline. Existing Early Warning Systems detect abnormal vital signs but do not verify whether appropriate escalation actions — such as timely antibiotics, senior review, or ICU referral — actually occurred.

The system detects deterioration.
It does not detect inaction.

Explainable GenAI for Escalation Intelligence

APPLICATION & PROPOSED METHOD

1 APPLICATION

Setting: General hospital wards.

Target Users: Ward doctors, nurses, and rapid response teams. The system continuously monitors physiological deterioration and verifies whether escalation actions occur within guideline-defined time windows. This functions as human-in-the-loop decision support.

2 PROPOSED METHOD

1. ML Model analyzes time-series vital signs and laboratory trends to detect sepsis-related physiological decline.

2. Rule-based Escalation Engine evaluates adherence to clinical guidelines (antibiotics within 1 hour, senior review, ICU referral).

3. GenAI Reasoning Layer analyzes the deterioration-to-action timeline and generates clear, human-readable explanations identifying missing actions. *Example Alert:* “Patient met sepsis criteria at 14:20. No antibiotics administered within 60 minutes. Escalation recommended.”

Validation, Data & Scalable Impact

DATASETS / DATA SOURCES

De-identified retrospective hospital ward data, structured EHR-style datasets, and synthetic patient timelines for controlled validation.

EXPERIMENTS & EVALUATION

Compare AI-detected failure-to-escalate cases with clinician-reviewed ground truth, evaluate precision and recall, measure time-to-escalation improvement, and assess explainability clarity and clinician trust.

NOVELTY

This solution detects the absence of appropriate action after deterioration, shifting AI from risk prediction to patient safety accountability, unlike most systems that only predict deterioration.

SCOPE TO SCALE

Extendable to respiratory failure, acute kidney injury, and post-operative deterioration, designed for integration into hospital clinical monitoring systems.

Designed for safe, explainable, workflow-integrated AI deployment in hospital wards.